Zhichao Wang

Curriculum Vitae
Email: zhw036@ucsd.edu
https://zhichao-w.github.io

EDUCATION

University of California San Diego, USA *PhD candidate in Mathematics, 2019-present*

- Advisors: Prof. Ioana Dumitriu and Prof. Todd Kemp

Texas A&M University, College Station, USA *MSc. in Mathematics, 2017-2019*

- Advisors: Prof. Michael Anshelevich

Texas A&M University, College Station, USA *Exchange Program, 2016-2017*

Beihang University, Beijing, China *BSc. in Mathematics and Applied Mathematics, 2013-2017*

RESEARCH INTERESTS

- High-dimensional Probability · Random Matrix Theory and its Applications · Free Probability Theory
- Deep Learning Theory · Feature Learning · Kernel Methods · Random Feature Regression
- High-Dimensional Statistics · Asymptotic Statistics

PUBLICATIONS AND PREPRINTS

- **High-Dimensional Asymptotics of Feature Learning in the Early Phase of Neural Network Training** (with Jimmy Ba, Murat A. Erdogdu, Taiji Suzuki, Denny Wu, and Greg Yang), in preparation.
- **Overparameterized random feature regression with nearly orthogonal data** (with Yizhe Zhu), Accepted by AISTATS 2023.
- **Spectral evolution and invariance in linear-width neural networks** (with Andrew Engel, Anand Sarwate, Ioana Dumitriu, Tony Chiang), submitted.
- **High-dimensional Asymptotics of Feature Learning: How One Gradient Step Improves the Representation** (with Jimmy Ba, Murat A. Erdogdu, Taiji Suzuki, Denny Wu, and Greg Yang), Accepted by NeurIPS 2022.
- **Deformed semicircle law and concentration of nonlinear random matrices for ultra-wide neural networks** (with Yizhe Zhu), submitted.
- **Tree convolution for probability distributions with unbounded support** (with Ethan Davis and David Jekel), Latin American Journal of Probability and Mathematics Statistics 18.2 (2021), pp. 1585-1623.
- **Spectra of the Conjugate Kernel and Neural Tangent Kernel for linear-width neural networks** (with Zhou Fan), Neural Information Processing Systems (NeurIPS), 2020, *Oral Presentation.*
- **Principal components in linear mixed models with general bulk** (with Zhou Fan and Yi Sun), The Annals of Statistics, 49.3 (2021), pp. 1489-1513.
- **Higher variations for free Lévy processes** (with Michael Anshelevich), Studia Math. 252 (2020), pp. 49-81.
- **Convergence of the Powers of Free Triangular Arrays to Higher Variations of Free Lévy Processes**, Master's thesis, Texas A & M University, *available electronically.*

WORK EXPERIENCES

Internship at Pacific Northwest National Laboratory *06/2021- 09/2022*

Advisor: Tony Chiang. Deep learning research and software development.

Teaching Assistant at Math Department of UC San Diego *09/2019-12/2021*

MATH 10B, MATH 3C, MATH 10C, MATH 180A, and MATH 280BC.

Teaching Assistant at Math Department of Texas A&M University *08/2018-07/2019*

MATH 251, MATH 308 and MATH 220.

Grader of Math Department at Texas A&M University *01/2017-05/2018*

MATH 467, MATH 411 and MATH 152.

Internship at AMSS, Chinese Academy of Sciences *07/2016-08/2016*

Advisor: Prof. Feimin Huang, AMSS, Chinese Academy of Sciences

Mentor at Math Department, Beihang University *10/2014-07/2015*

HONORS AND AWARDS

2019	Math Fellowship, UCSD, USA
2017	Math Fellowship, TAMU, USA
2016, 2017	Study Abroad Scholarships, Beihang University, China
2016	Meritorious Winner of MCM/ICM Contest, USA
2016	Second Prize in the 26th Fengru Cup Competition of Academic Research Works, China
2015, 2016	Hua Luogeng Scholarships, AMSS, Chinese Academy of Sciences
2014, 2015	Huatong Scholarships of SMSS, Beihang University, China
2014, 2015	Scholarships of Academic Performance, Beihang University, China
2014	First Prize in College Students Physics Contest in Beijing, China
2013	Freshman Scholarship, Beihang University, China

TALKS

06/2022	RMMC Summer School at University of Wyoming.
03/2022	Combinatorics and Probability Seminar at UC Irvine.
12/2021	Frontier Probability Days 2021, at Las Vegas.
09/2021	Universality and Integrability in Random Matrix Theory and Interacting Particle Systems Workshop, at MSRI (virtual).
03/2021	Machine Learning Seminar, at Pacific Northwest National Laboratory (virtual).
12/2020	Neural Information Processing Systems (NeurIPS) virtual oral presentation.
03/2020	Student Probability Seminar, at UCSD.
11/2019	Student Probability Seminar, at UCSD.
05/2017	Capstone Project Presentation, TAMU.
04/2016	Complex Dynamic Systems Seminar, Beihang.

SKILLS

Language	Chinese (Native), English (Fluent)
Software	Python, Matlab, LaTeX, Tensorflow, Keras, PyTorch, JAX, C++

PROFESSIONAL SERVICE

Journal reviewer: Annals of Statistics, Canadian Journal of Mathematics.
Conference reviewer: AISTATS '22, AISTATS '23.